



Semi-synthetic Hyaluronic Acid Hydrogels Promote Muscle Regeneration by Inducing BAT Differentiation of FAPs

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INTRODUCTION

Fibro-adipogenic progenitor cells (FAPs) have been shown to be key regulators in repair of injured skeletal muscles. In uninjured muscles, FAPs are typically present in a quiescent state; however, they rapidly expand in response to muscle injury and can adopt a brown-adipose tissue (BAT) phenotype to support satellite cell myogenesis [1].

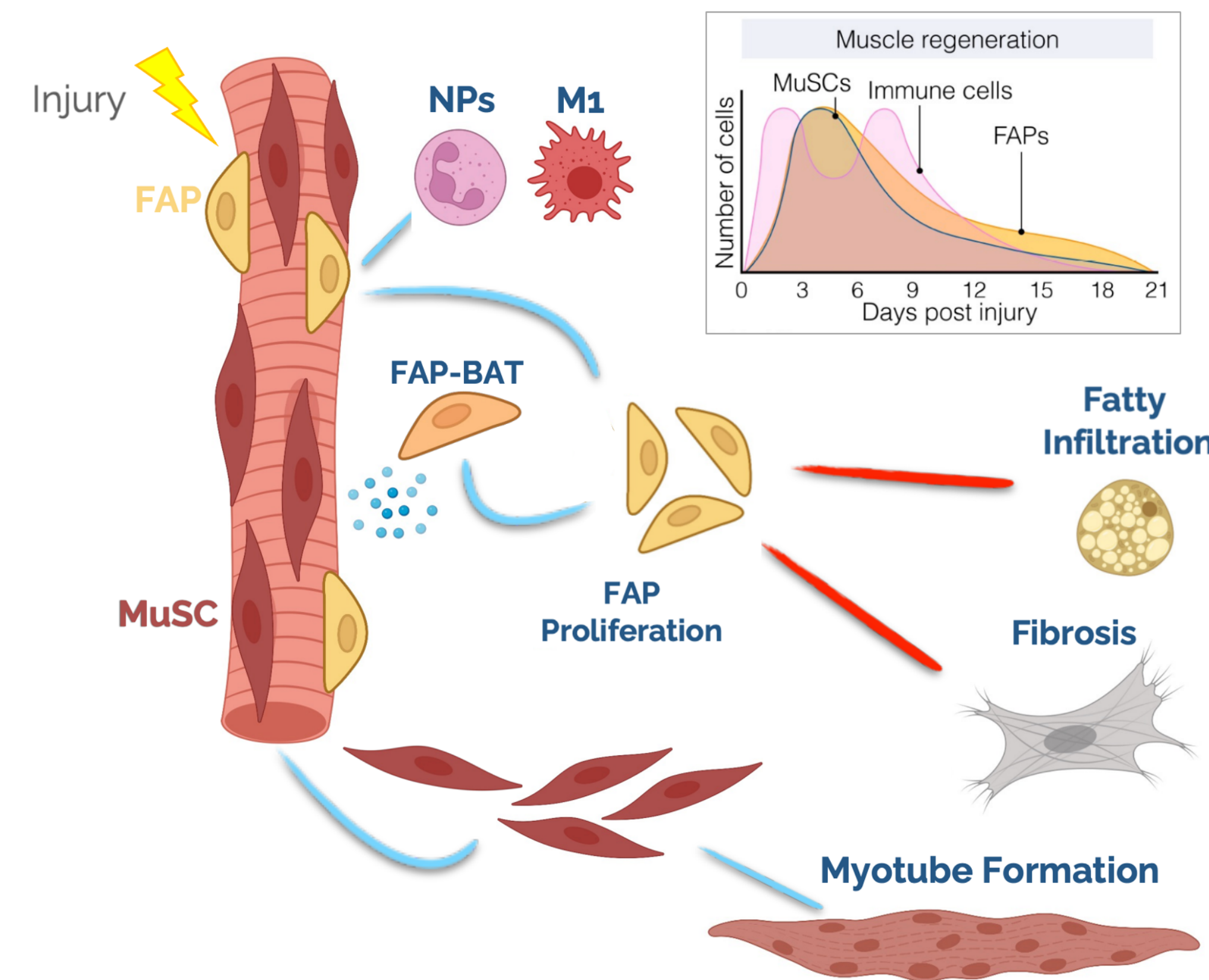


Figure 1. Schematic of cellular response after muscle injury. Inset: time course of cellular recruitment after muscle injury. Adapted from [2].

APPROACH

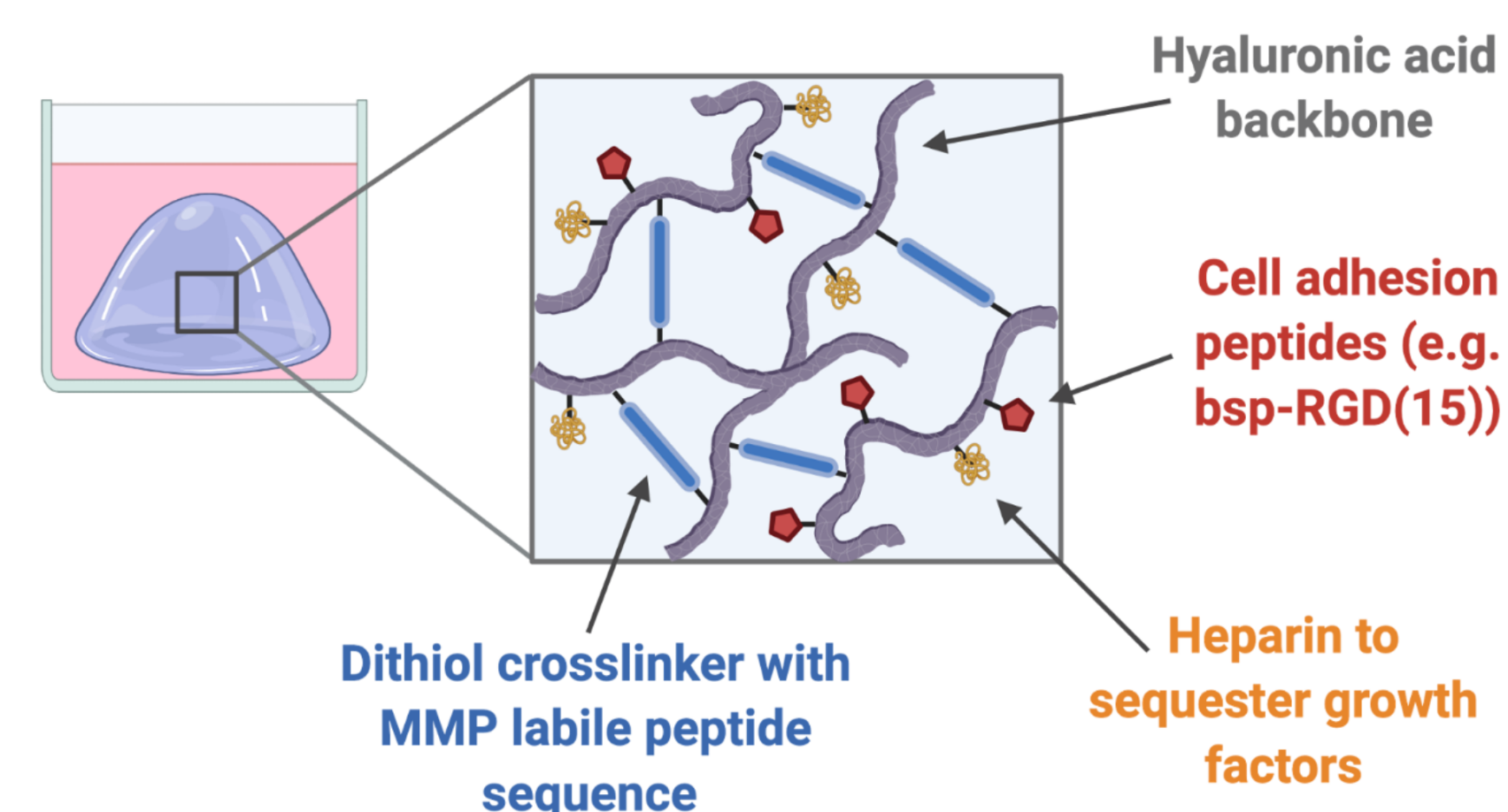


Figure 2. Semi-synthetic hyaluronic acid (HyA) hydrogel for tissue regeneration and stem cell transplantation.

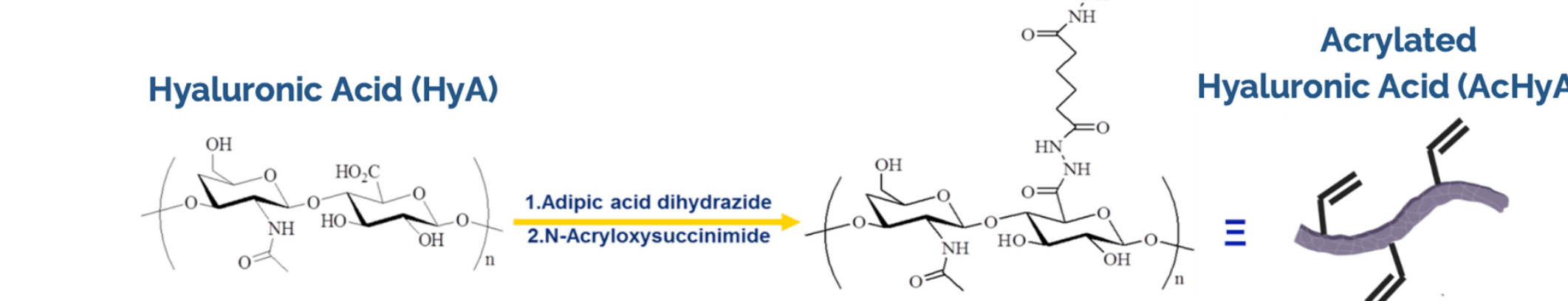
Highly tunable:

- Mechanical properties
- Degradation rate
- Cell specificity
- Growth factor sequestration

We hypothesize that our novel hydrogel would promote beige adipogenesis of human FAPs and thereby induce muscle regeneration.

METHODS

Synthesis of AcHyA



Synthesis of HyA bioconjugates

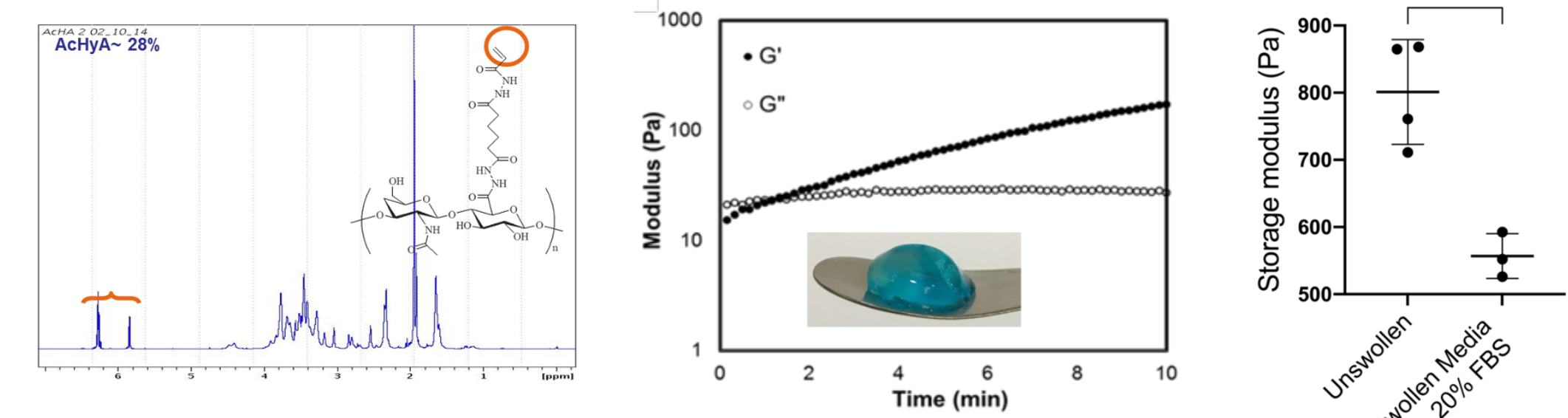
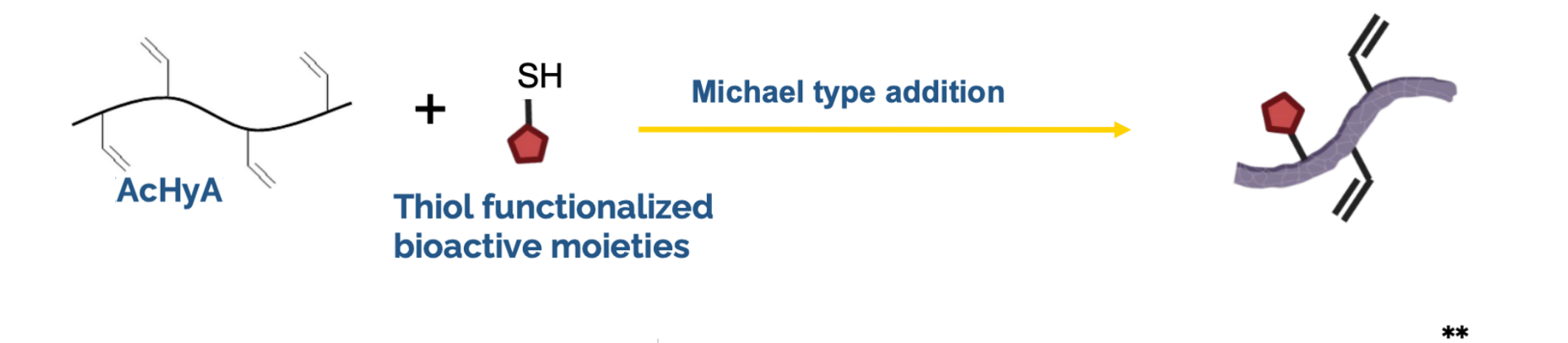


Figure 3. HyA macromer acrylation and synthesis of bioconjugates. The HyA hydrogels were synthesized using previously reported methods [3,4]. Briefly, HyA derivative carrying hydrazide groups (HyA-ADH) was synthesized and N-acryloxysuccinimide was subsequently reacted with the HyA-ADH solution to generate acrylate groups on the HyA (AcHyA). For growth factor sequestration, SH-heparin was also synthesized to bind AcHyA via Michael-type addition.

Deltoid muscle biopsies from RC tear patients

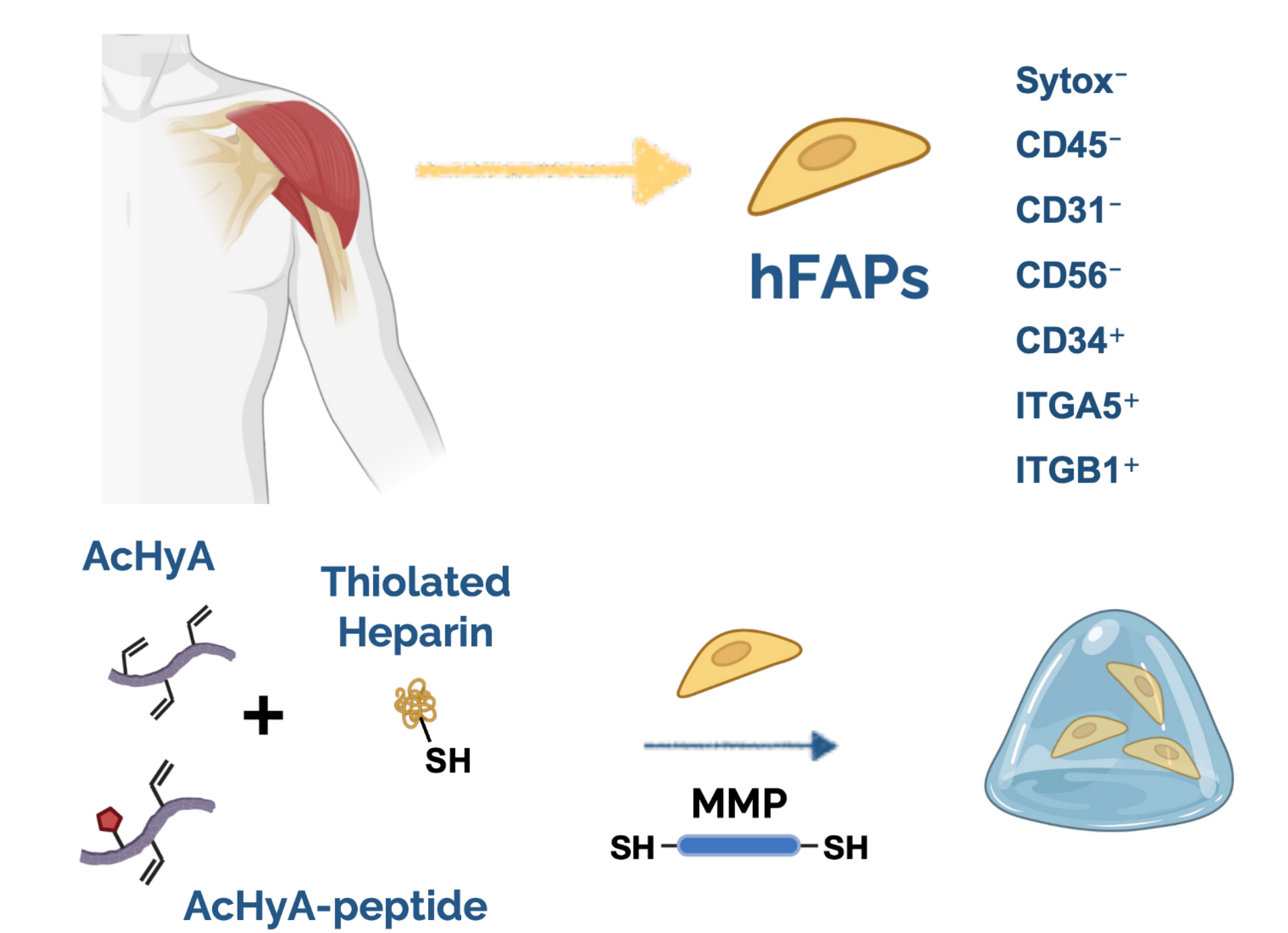


Figure 4. hFAP harvest and encapsulation. hFAPs were harvested from deltoid muscle biopsies (10 mg samples) performed on patients undergoing arthroscopic rotator cuff repair. Cells were encapsulated by mixing the cells into the macromer precursor solution along with one of four cell binding peptides (RGD, C16, Ag73, and P3). Gelation was initiated by addition of a bis-cysteine terminated peptide-cleavable crosslinker.

IN VITRO RESULTS

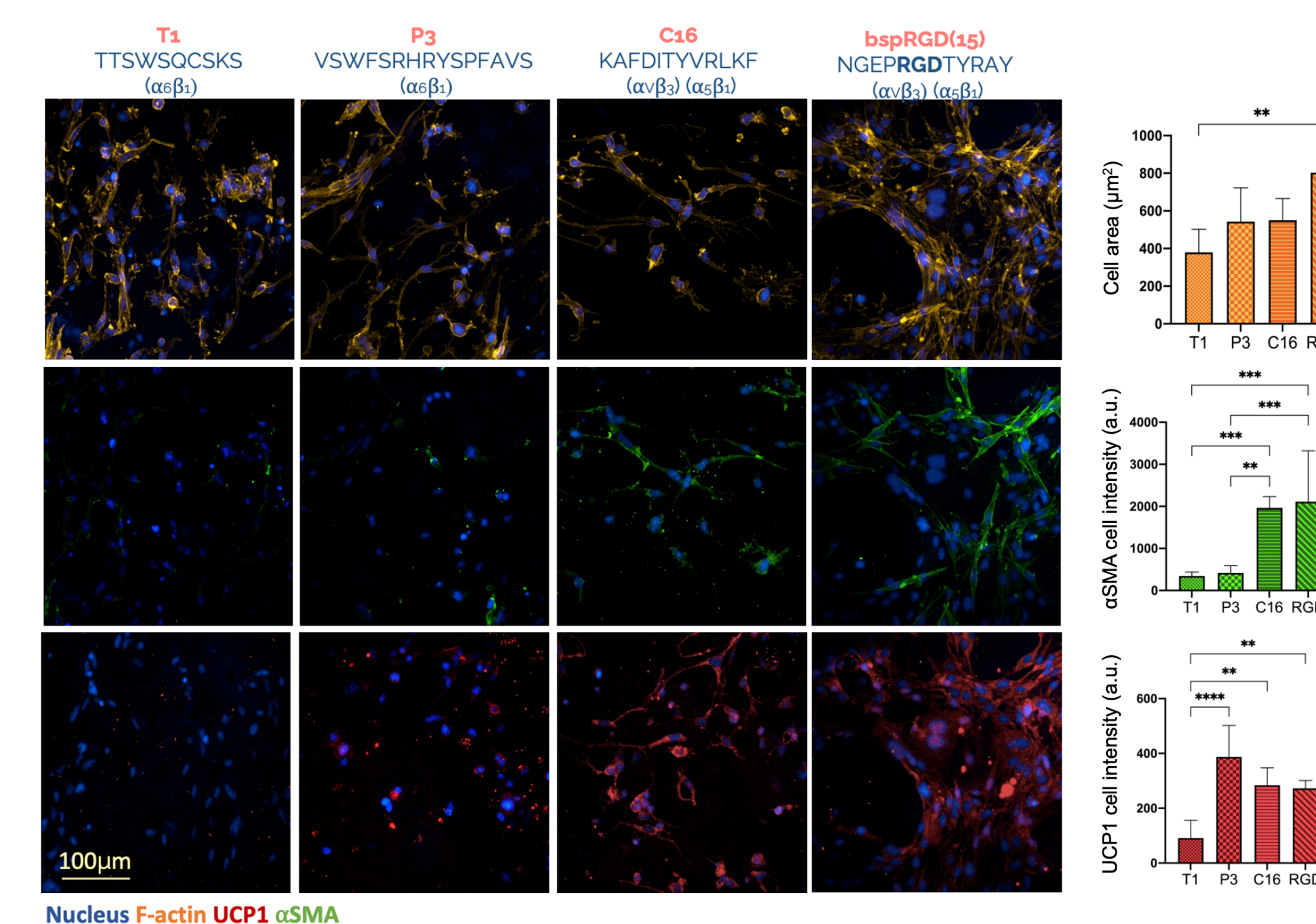


Figure 5. Culture of hFAPs in the HyA hydrogels with various integrin engaging peptides for 7 days. There was increased expression of the BAT marker UCP1 in the HyA-bsp-RGD(15), C16, and P3 peptide hydrogels compared to HyA-T1. α SMA, a fibrotic marker, was elevated in the HyA-bsp-RGD(15) and -C16 peptide hydrogels.

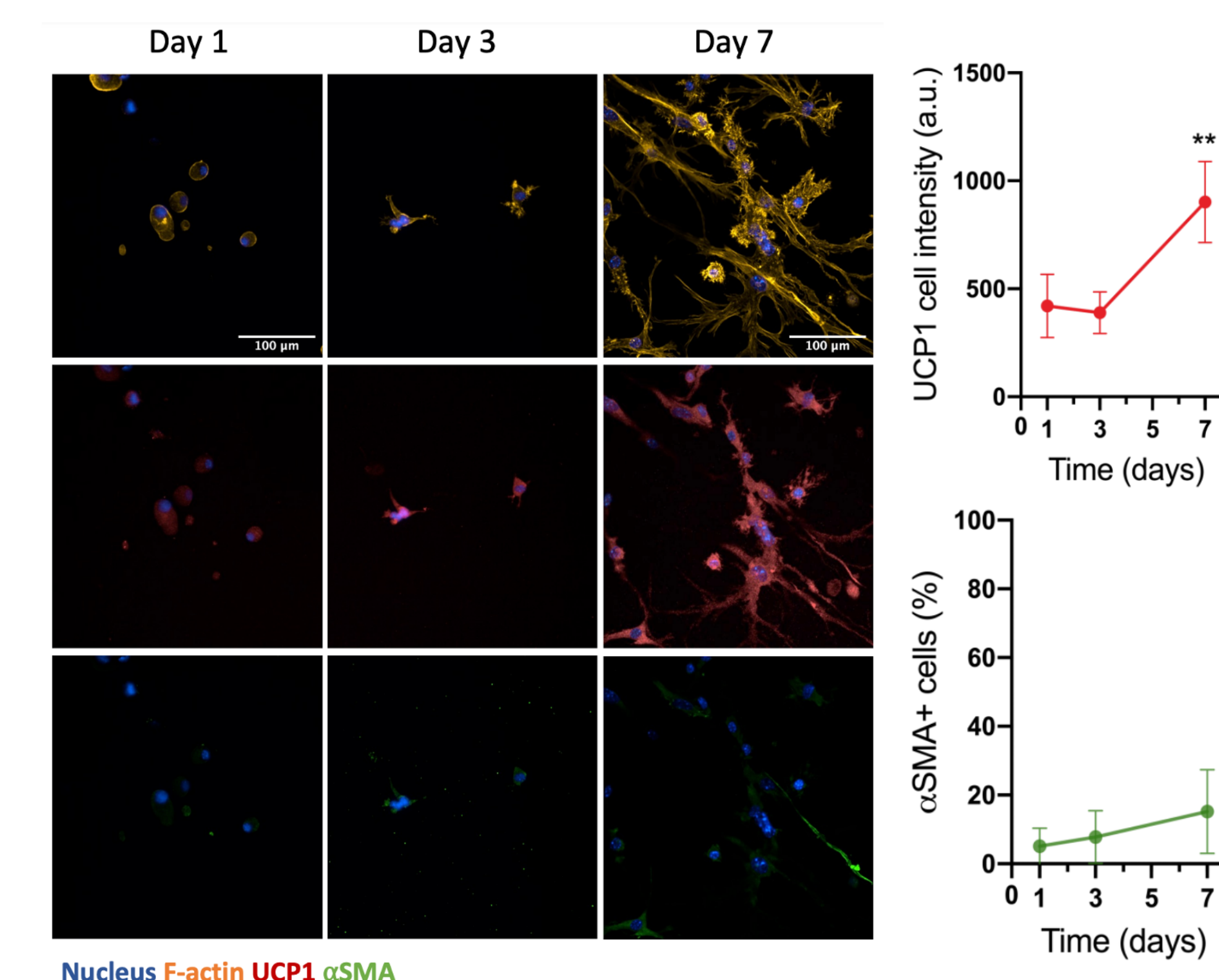


Figure 6. Time course of BAT phenotype. Immunostaining of hFAPs for UCP1 and α SMA after 1, 3 and 7 days in the HyA hydrogel with bsp-RGD(15). There was increased expression of the BAT marker UCP1, compared to day 1 and day 3. In contrast, α SMA, a fibrotic marker, did not increase significantly over time.

IN VIVO RESULTS

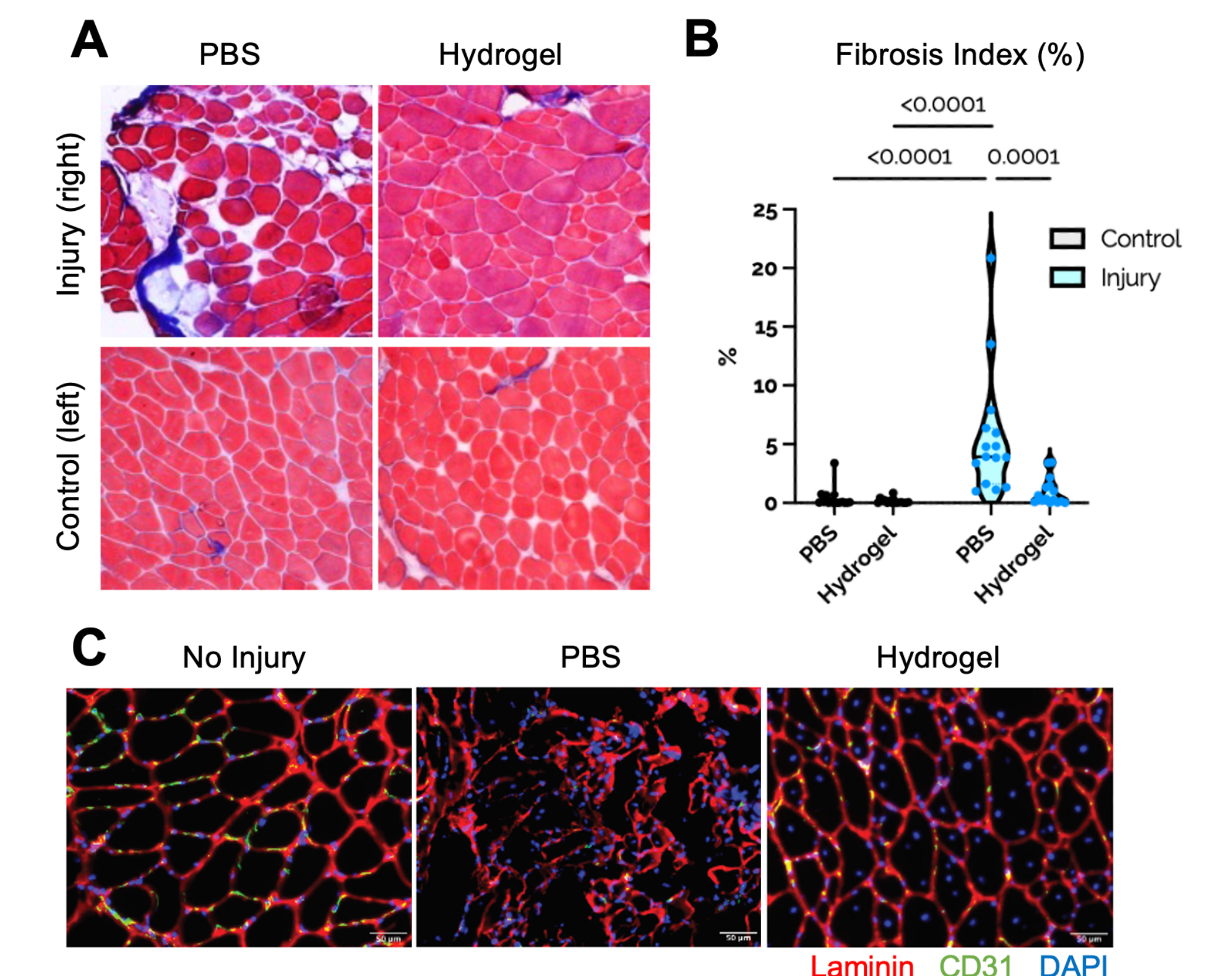


Figure 7. HyA hydrogels promote muscle regeneration in vivo. (A) Masson's Trichrome Staining of experimental groups. Blue areas represent areas of fibrosis. (B) Quantified fibrosis index (%) of treatment and control groups. (C) Assessment of myofiber cross-sectional area and vascularization via laminin and CD31 immunofluorescence staining, respectively. Paired muscle weight loss was reduced in HyA treatment group compared to PBS.

CONCLUSIONS

Our results suggest that hydrogel-driven FAP-BAT differentiation may contribute to the positive regenerative outcomes after treatment of muscular injury with our hydrogel, highlighting the potential for this material to serve as an effective therapeutic for rotator cuff diseases. Additional characterization of encapsulated hFAPs to verify BAT phenotype and promyogenic activity (flow cytometry, cytokine secretion, RNA-seq) are necessitated to elucidate the precise mechanism of FAP-facilitated muscle repair in our HyA hydrogels.

ACKNOWLEDGEMENTS

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